Course Specification
(Bachelor)

Course Title:	Biostatistics
Course Code:	MEDI – 1201
Program:	PharmD
Department:	Pharmaceutical Sciences
College:	College of Pharmacy
Institution:	Shaqra University
Version:	2
Last Revision Date:	16-01-2025

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A. General information about the course:

1. Course Identification

1. Credit hours: 2 (2+0)					
2. Course type					
A.	☐University	☒ College	☐Department	☐Track	☐Others
B.	☒ Required		☐Elective		
3. Level/year at which this course is offered: (5th Level / 3rd Year)					
4. Course General Description:					
Presents fundamental concepts in applied probability, exploratory data analysis, and statistical inference, focusing on probability and analysis of one and two samples. Topics include discrete and continuous probability models; expectation and variance; central limit theorem; inference, including hypothesis testing and confidence for means, proportions, and counts; maximum likelihood estimation; sample size determinations; elementary non-parametric methods; graphical displays; and data transformations.					
5. Pre-requirements for this course (if any)**:**					
N/A					
6. Co-requisites for this course (if any)**:**					
N/A					
7. Course Main Objective(s):					
• To understand the use of statistics in research papers in general public health setting of wet lab and clinical research. • The students should be able to do simple estimation and to conduct simple hypothesis testing					

2. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1	Traditional classroom	2	100%
2	E-learning		
3	Hybrid • Traditional classroom • E-learning		
4	Distance learning		

3. Contact Hours (based on the academic semester)

No	Activity	Contact Hours
1.	**Lectures**	30
2.	**Laboratory/Studio**	
3.	**Field**	
4.	**Tutorial**	
5.	**Others (specify)**	
Total		**30**

B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of PLOs aligned with program	Teaching Strategies	Assessment Methods
1.0	**Knowledge and understanding**			
1.1	Describe the basics of statistics used in research	K1	• Lectures • Class discussion	Mid-term and final exams
1.2	Describe the concept and principle in estimating and hypothesis testing using suitable mathematical expressions.	K2	• Lectures • Class discussion	Mid-term and final exams
2.0	**Skills**			
2.1	Utilize application of statistical techniques in pharmaceutical sciences.	S1	• Lectures • Class discussion	Mid-term and final exams
2.2	Evaluate scientific and professional literature critically to be utilized in evidence-based practice and problem solving using statistical techniques.	S2	• Lectures • Class discussion • Problem solving exercise	Mid-term and final exam, Quizzes, Evaluation of exercise

Code	Course Learning Outcomes	Code of PLOs aligned with program	Teaching Strategies	Assessment Methods
3.0	**Values, autonomy, and responsibility**			
3.1	Express leadership skills professional ethics, integrity, attitudes, accountability, and acceptance of responsibility within team in various settings.	V1	Presentation, group discussions/Assignments	Observational Evaluation/ Rubric assessment by faculty

C. Course Content

No.	List of Topics	Contact Hours
1.	What are statistics: Introduction to the basic terms	2
2.	Frequency distribution: Cumulative and relative frequency, Measures of Shape: Normal Distribution. Skewness: Positive Skew, Negative Skew	2
3.	Descriptive Statistics- Measures of Central Tendency: Mean, Sample mean, Population mean, Mode, Median.	2
4.	Measures of dispersion: Range, Quartiles, IQR, Variance	2
5.	Standard Deviation, Standard Error, Z score Outlier calculation	2
6.	Basic probability theory	2
7.	Central limit theorem and its application	2
8.	Hypothesis testing problems: Constructing Hypothesis, Concepts of Hypothesis Testing: Null and Alternative Hypothesis	2
9.	Level of Significance, P value, Chances of error, confidence level, Types of errors: Type I and Type II errors, Power of a test	4
10.	Directional Tests: One tail Test (Left Tail), One tail Test (Right Tail), Two tailed test, acceptance region, Rejection region and Critical value Z test	2
11.	Parametric Tests: Students “t” Test – Unpaired and Paired “t” test, Chi Square Test, ANOVA, Post hoc test	2
12.	Non – Parametric Tests: Kruskal –Wallis H test	2

13.	Wilcoxon's Signed Rank Test and Mann Whitney Test, Contingency tables: odds ratio	2
14.	Linear correlation and linear regression	2
Total		30

D. Students Assessment Activities

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.			
2.			
3.			
...			

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.).

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	• Elementary Statistics, 11th Ed, Johnson and Kuby, ISBN-13:978-0538733502 • Elementary Statistics, 8th Ed, Bluman, ISBN13:9780073386102 • Biostatistics: Basic Concepts and Methodology for the Health Sciences, 10th Edition, Wayne W. Daniel, Chad L. Cross International Student Version, Wiley Publication, ISBN:798-1-118-65291-6
Supportive References	• Introductory Statistics. Wonnacott, T. H., and Wonnacott, R. J. John Wiley & Sons. 1969, ISBN-13:978-0471615187
Electronic Materials	• https://stats.blue/ • https://online.stat.psu.edu/statprogram/tutorials/statist ical-software • https://datatab.net/ • https://www.socscistatistics.com/pvalues/ • https://www.graphpad.com/quickcalcs/pvalue1.cfm
Other Learning Materials	• Multimedia associated with the textbook and the relevant websites and SPSS, STATA, EXCEL

2. Required Facilities and equipment

Items	Resources
facilities (Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)	• Well-equipped lecture room with at least 30-40 seats
Technology equipment (projector, smart board, software)	• Data show, Smartboard, Power point, Video conferencing software.
Other equipment (depending on the nature of the specialty)	N/A

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Students	Indirect (Surveys)
Effectiveness of Students' assessment	Faculty, Program coordinator	Direct
Quality of learning resources	Students	Direct
The extent to which CLOs have been achieved	Quality and development department	Indirect (Surveys)

Assessors (Students, Faculty, Program Leaders, Peer Reviewers, Others (specify)
Assessment Methods (Direct, Indirect)

G. Specification Approval

COUNCIL /COMMITTEE	
REFERENCE NO.	
DATE	